

Square, Square roots, Cube roots,
Multiplication table tricks

Square Calculation Technique (UP to 50)

Last digit 5

$$35^2 = \boxed{12} \boxed{25} \rightarrow 5^2 = 25$$

$\rightarrow 3 \times (3+1) = 12$

Note:- Must Remember square from 1 to 50

Base Method ($50^2 = 2500$) $\Rightarrow$ For 26^2 to 75^2

Steps

- i) Check range b/w 26^2 to 75^2
- ii) Either $\oplus / \ominus$ from 50
- iii) Add or subtract difference to 25.

i) Positive Case

$$\begin{array}{c} \textcircled{+4}^2 \quad 54^2 = \textcircled{29} \textcircled{\underline{16}} \\ \downarrow \quad \nearrow \quad \nearrow \\ \text{Base} = 25 + 4 \end{array}$$

$$\begin{array}{c} \textcircled{+2}^2 \quad 52^2 = \textcircled{27} \textcircled{\underline{04}} \\ \downarrow \quad \nearrow \quad \nearrow \quad \downarrow \\ \text{Base} = 25 + 2 \quad (04, \text{ because we need two digits}) \end{array}$$

Remainder Case

$$\begin{array}{c} \textcircled{+11} \quad 61^2 = \textcircled{37} \textcircled{\underline{21}} \quad (11^2 = 121; \therefore \text{ we need two digits only.} \\ \downarrow \quad \nearrow \quad \nearrow \quad \therefore \text{ treat } +1 \text{ as remainder}) \\ \text{Base} = 25 + 11 + 1 (\text{remainder}) \end{array}$$

ii) Negative case

$$\textcircled{-8} \quad 42^2 = 1764$$

$\swarrow \quad \searrow$
 $25-8 \quad 8^2$

$$\textcircled{-3} \quad 47^2 = 2209$$

$\swarrow \quad \searrow$
 $25-3 \quad 3^2$

Remainder Case $+2$

$$\textcircled{-16} \quad 34^2 = 1156$$

$\swarrow \quad \searrow$
 $25-16+2 \quad 16^2$

Base ($100^2 = 10000$) $\Rightarrow$ For 76^2 to 150^2

Steps

i) check range b/w 76^2 to 150^2

ii) Either $\oplus / \ominus$ from 100

iii) Add or subtract difference from number itself

$$\textcircled{+4} (104)^2 = 108 \ 16$$

$$\downarrow \quad \downarrow$$

$$104+4 \quad 4^2$$

$$\textcircled{-11} 89^2 = 7921 \textcircled{+1}$$

$$\swarrow \quad \downarrow$$

$$89-11+1 \quad 11^2$$

$$\textcircled{-7} (93)^2 = 86 \ 49$$

$$\downarrow \quad \downarrow$$

$$93-7 \quad 7^2$$

$$\textcircled{+12} (112)^2 = 125 \ 44 \textcircled{+1}$$

$$\swarrow \quad \downarrow$$

$$112+12+1 \quad 12^2$$

Square root

Base $\Rightarrow 50$ (Number < 5600)

$$\sqrt{1369} = \textcircled{50} - \textcircled{13} = 37$$

↓
25 की नीचे की
एक square
है!

↓

$$13^2 = 1\textcircled{69}$$

$$\sqrt{1521} = 50 - 11 = 39$$

→ If Number < 2500

$$\sqrt{3136} = 50 + 6 = 56$$

→ If Number > 2500

$$\sqrt{2401} = 50 - 1 = 49$$

Base = 100 $\Rightarrow$ (For numbers 5600 to 15,500)

$\sqrt{5776} = 100 + \underbrace{(24)}_{\substack{\rightarrow \text{For } < 10000 \\ \downarrow \\ 24^2 = 576}}$

$\sqrt{11449} = 100 + 7 = \underbrace{107}_{\substack{\rightarrow \text{For } > 10000 \\ \downarrow \\ 7^2 = 49}}$

Approx imperfect root

$$\sqrt{5} = \sqrt{4+1} \overset{\text{as it is}}{=} 2 + \frac{1}{4}$$

Diagram illustrating the approximation of $\sqrt{5}$. The expression $\sqrt{5} = \sqrt{4+1}$ is shown. A bracket under $\sqrt{4}$ is labeled "as it is". The result is $2 + \frac{1}{4}$. A curved arrow points from the 4 in the denominator to the 2 in the integer part, labeled "after root". A straight arrow points from the 1 in the numerator to the 4 in the denominator, labeled "x2".

$$= 2 + 0.25 = 2.25$$

$$\sqrt{8} = \sqrt{9-1} = 3 - \frac{1}{6}$$

Diagram illustrating the approximation of $\sqrt{8}$. The expression $\sqrt{8} = \sqrt{9-1}$ is shown. A bracket under $\sqrt{9}$ is labeled "as it is". The result is $3 - \frac{1}{6}$. A curved arrow points from the 9 in the denominator to the 3 in the integer part, labeled "after root". A straight arrow points from the 1 in the numerator to the 6 in the denominator, labeled "x2".

$$3 - 0.167 = 2.833$$

$$\sqrt{18} = \sqrt{16+2} = 4 + \frac{2}{8} = 4 + 0.25 = 4.25$$

Diagram illustrating the binomial expansion for $\sqrt{18}$. The expression is written as $\sqrt{16+2}$. A bracket under 16 is labeled 4, and a bracket under 2 is labeled 8. An arrow points from the 4 to the 2, and another arrow points from the 2 to the 8, indicating the relationship $4 \times 2 = 8$.

$$\sqrt{46} = \sqrt{49-3} = 7 - \frac{3}{14} = 7 - 0.214 = 6.786$$

Universal Square

$$\begin{array}{r} 25^2 \\ \downarrow \quad \downarrow \\ 22 \quad 52 \\ 4 \quad 25 \\ +2 \quad 0 \times \\ \hline 625 \end{array} \quad \begin{array}{l} 2 \times 5 \times 2 \end{array}$$

Universal Square root

$$60 \mid 84$$

x



$$2^2 \text{ or } 8^2$$

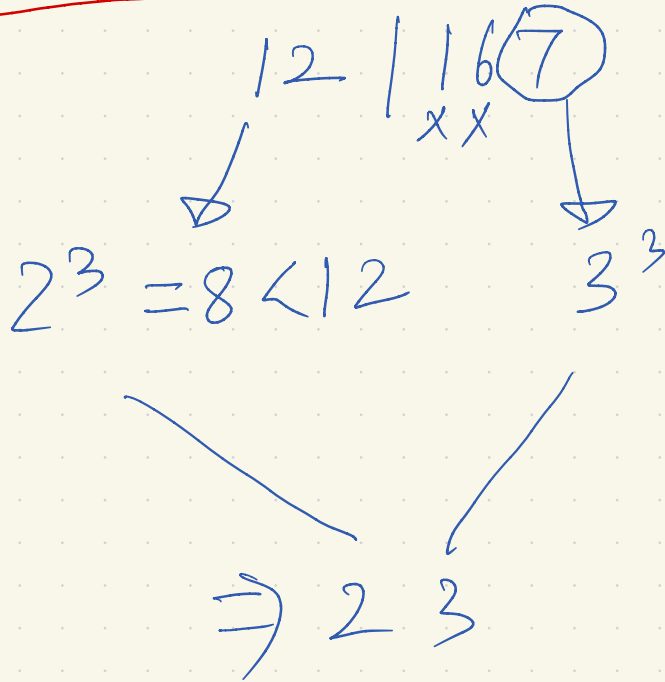


$$7^2 = 49 < 60$$

$$\Rightarrow 72 \mid 78$$

$$\text{Range} \Rightarrow \underbrace{4900}_{\neq 72 \text{ (far)}} - 6084 - \underbrace{6400}_{\text{Near } 78} \quad (70^2) \quad (80^2)$$

Cube root



Multiplication Table Method (1 to 100)

Example:-

2	6	$\rightarrow$ $\textcircled{+1}$ $42 = 52$
4	12	$\rightarrow$ $42 = 52$
6	18	$\rightarrow$ $(6+1)8 = 78$
8	24	$\rightarrow$ $(8+2)4 = 104$
10	30	$\rightarrow$ $(10+3)0 = 130$
12	36	$\rightarrow$ $(12+3)6 = 156$
14	42	$\rightarrow$ $(14+4)2 = 182$
16	48	$\rightarrow$ $(16+4)8 = 208$
18	54	$\rightarrow$ $(18+5)4 = 234$
20	60	$\rightarrow$ $(20+6)0 = 260$